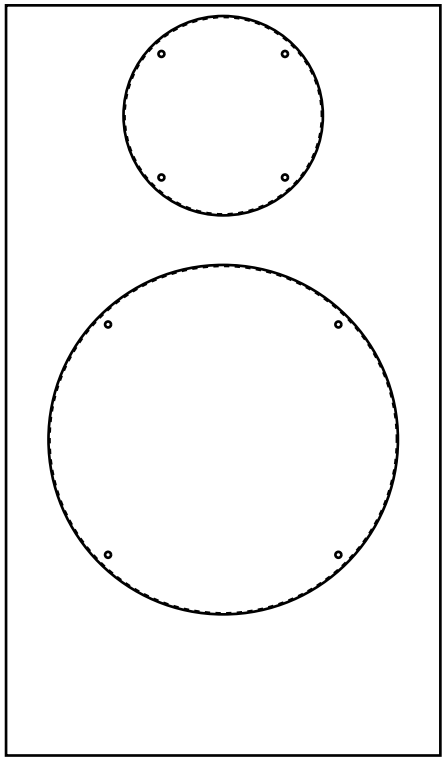
01 / One box, matching manufacturing sets

220 W x 380 H x 262.652 D | 10 mm panels | 16.100 L nominal net

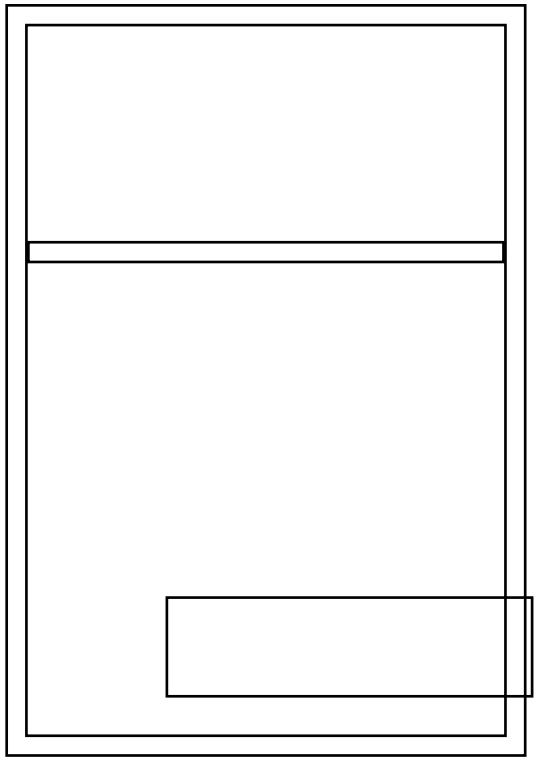
FRONT



Centres: z160 / z324.1

177 / 101 acrylic through-holes

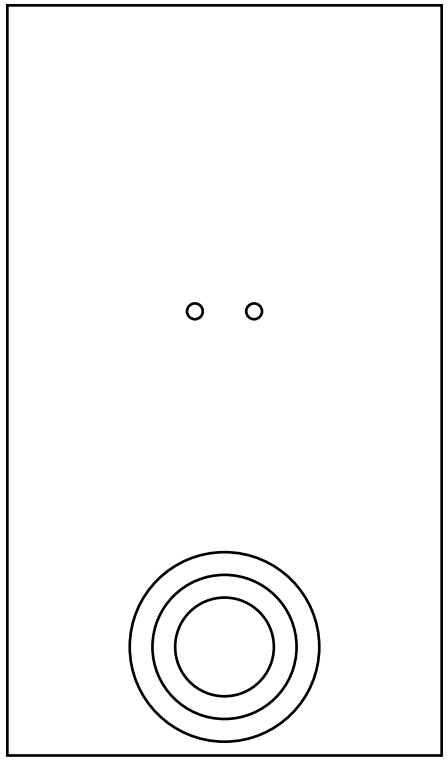
SIDE SECTION



Brace underside z250, thickness10

Port axis z55; 50 ID x 185 assembled

REAR



Rear removes with port + crossover

Two 8 mm binding-post openings

Driver seats are in printed adapters: woofer depth3.6, tweeter3.0; 165 / 88.5 PCD at 45 + 90n degrees.

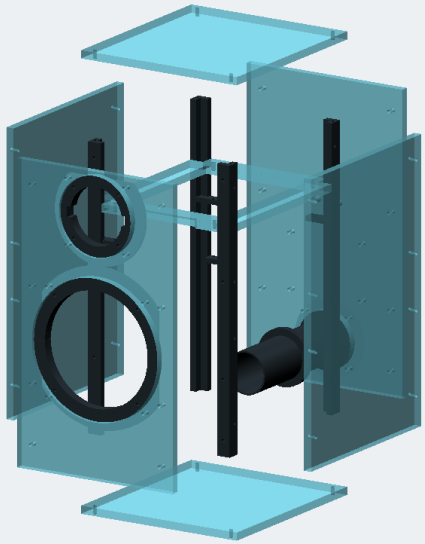
Circular acrylic openings stay simple. Printed tweeter terminal relief: 81.8 x 13; centre bore73.

Dimensioned profiles are in DXF/SVG. Cabinet depth excludes the port flange, driver protrusions and screw heads.

02 / Seven sheets + thirteen printed fittings

Unrestricted acrylic build / 20 manufactured pieces / 74 M3 screws

UNRESTRICTED / LASER-CUT BUILD
Six outer sheets + one window brace Eight rails + three adapter pieces + two port pieces



FRONT 220.000 x 380.000 x 10

REAR 220.000 x 380.000 x 10

LEFT 241.652 x 359.000 x 10

RIGHT 241.652 x 359.000 x 10

BOTTOM 220.000 x 241.652 x 10

TOP 220.000 x 241.652 x 10

BRACE 199.000 x 240.652 x 10

8 rails, 179.5 long each

2 woofer adapter halves

1 tweeter adapter

2 port pieces

All outer panels use plain butt joints and 0.5 mm compressed gasket gaps.

Printed rails carry M3 inserts. Upper rails integrate the four brace shelves; no separate shelf brackets.

Brace has 30 mm frame, corner rail cutouts and a shallow front notch to clear the woofer carrier.

Use all seven whole-panel DXFs OR SVGs, one each. Whole-panel STLs reproduce exactly those 10 mm profiles.

Test actual thickness and kerf with the supplied 80 x 50 fit coupon before cutting these panels.

The top tweeter ligament is only 5.4 mm: protect during handling and fit its supporting carrier.

03 / A1 mini: identical size, prototype-only seams

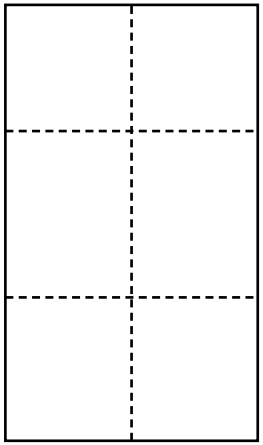
32 sheet tiles + 13 fittings + 15 joining pieces = 60 prints / 47 unique STL files

Front 6, rear 6, left 4, right 4, top 4, bottom 4, brace 4.

Side tiles print upright (179.25 high); other sheet tiles print flat. Saved STLs are already oriented.

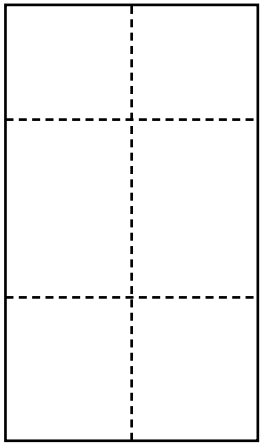
Right matches left subdivision; bottom matches top. Dashed lines are 0.5 mm gasketed prototype seams.

FRONT



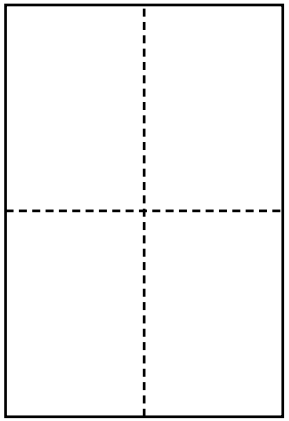
2 x 3 tiles

REAR



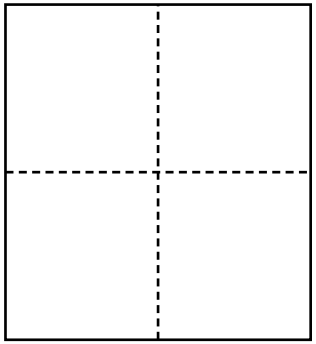
2 x 3 tiles

LEFT



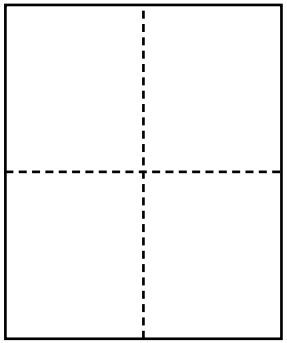
2 x 2 tiles

TOP



2 x 2 tiles

BRACE



2 x 2 tiles

Print splice_strip.stl 9 times and splice_cross.stl 6 times; every other STL once.

All outer-panel splices are external. Four brace strips lie on its upper face. Extra holes are prototype-only.

Use the A1 front rails: they contain eight additional insert pilots for the front-edge strips.

Tile DXFs/SVGs match the tile STLs, not the unsplit full sheet. Exact offsets are in prototype-tile-map.csv.

Never scale to fit. Check brims, supports and purge paths in the slicer; the side tiles are tall, narrow prints.

A construction prototype does not establish acrylic panel stiffness, damping or acoustic performance.

04 / M3 joints, seals and assembly access

No countersinks or threads in acrylic / rear panel is the service panel

PRODUCTION HARDWARE	JOINT ASSUMPTIONS
58 x M3-16, 8 x M3-10	Acrylic holes4.5 / insert pilots4.2
4 x M3-20, 4 x M3-6	Flat washer + cushion stack1.5
60 inserts, M3 x4 long	M3 insert engagement checked in CAD
6 M3 nuts; 4 M3 x20 spacers	Compressed sheet / port gasket0.5
	Adapter radial clearance0.5
PROTOTYPE ADDITIONS	Woofer adapter split0.2
42 x M3-20, 8 inserts, 34 nuts	Driver gaskets thin and continuous
9 strips + 6 crosses	Check actual hardware before assembly.
See hardware list for washer counts.	

1. Fit inserts in printed parts. Join prototype tiles first if using the A1 set.
2. Assemble bottom, sides and rails. Lower the brace onto shelves and fasten it from above.
3. Fit top, square the box, then fit the front panel with its driver carriers. Brace must go in first.
4. Fit rear port, crossover/spacers and binding posts separately; connect leads with service slack.
5. Close the rear with eight perimeter screws. Seal corners, screw holes and every gasket intersection.
6. Dry-fit, leak-test and measure. Rear service: remove eight perimeter screws and withdraw backwards.

Detailed seal routes, port gaskets and adapter assembly are in assembly-instructions.md.

05 / Volume and validation

Unrestricted acrylic 16.100 L / A1 construction prototype 16.092 L nominal

gross cavity and driver wells	17.606772 L	DIGITALLY CHECKED
printed fittings and acrylic brace	0.738963 L	Manifold meshes / single CAD solids
port duct air	0.337939 L	A1 envelope fit in supplied orientations
woofer estimate	0.192103 L	2D profile / 10 mm extrusion agreement
tweeter estimate	0.103732 L	Screw, head, washer and insert clearances
crossover material estimate	0.120000 L	No part or component-envelope collisions
spacers	0.001536 L	Port fit and clear inlet approach
wiring allowance	0.002500 L	Minimum rectangular tile counts
hardware allowance	0.010000 L	84 native shapes checked against STEP
nominal net	16.100000 L	

Volume uses estimated driver, electronics and hardware displacement; installed net volume will differ.

The flat 220 mm baffle differs from the original tapered 180-220 mm, 6 degree tilted baffle.

Diffraction, crossover summation and panel resonances may change. Retain the original crossover initially.

Port: original nominal37 Hz; simplified cylinder estimate40.5 Hz. Neither validates this cabinet tuning.

Outstanding: physical fit, gasket compression, insert strength, slicing, support removal, leakage and sound.

Follow the included manufacturer references and acrylic fabrication guidance before production cutting.